

4: Exchange and Transport

Edexcel B A-level Biology (9BI0), Topic 4. This topic explains why larger organisms cannot rely on diffusion alone, how substances cross cell membranes, how insects, fish, mammals and flowering plants exchange gases, and how the mammalian circulatory system and the vascular tissue of plants move materials in bulk. Core practicals 5, 6, 7 and 8 belong to this topic.

4.1 Surface Area to Volume Ratio

4.1(i) How SA:V Affects Transport

Every cell must take in oxygen and nutrients and get rid of carbon dioxide and other wastes across its surface. The amount a cell *needs* depends on its volume (the amount of respiring cytoplasm), but the amount it can *exchange* depends on its surface area. As an object gets bigger, its volume increases faster than its surface area, so its surface area to volume ratio (SA:V) falls.

Side of cube (cm)	Surface area (cm ²)	Volume (cm ³)	SA:V
1	6	1	6 : 1
2	24	8	3 : 1
4	96	64	1.5 : 1
10	600	1000	0.6 : 1

For a cube of side L , $SA:V = 6L^2 / L^3 = 6 / L$, so doubling the side halves the ratio. The same idea applies to other shapes: for a sphere of radius r , $SA = 4\pi r^2$ and $V = (4/3)\pi r^3$, giving $SA:V = 3 / r$. For a cylinder of radius r and length h , $SA = 2\pi r^2 + 2\pi rh$ and $V = \pi r^2 h$. You may be asked to calculate these, so learn the formulae.

A second problem is distance. Diffusion is fast over a few micrometres but very slow over millimetres or centimetres, because the time taken rises with the square of the distance. The centre of a large organism would be starved of oxygen long before diffusion from the surface could supply it.

Fick's law: The rate of diffusion across a surface is proportional to (surface area × difference in concentration) ÷ thickness of the surface. Efficient exchange surfaces therefore have a large area, a steep concentration gradient and a short diffusion distance.

$$\text{rate of diffusion} \propto (\text{surface area} \times \text{concentration difference}) / \text{diffusion distance}$$

4.1(ii) Why Larger Organisms Need Mass Transport and Specialised Exchange Surfaces

Single-celled organisms such as *Amoeba* have a large SA:V and short distances, so diffusion across the cell surface meets their needs. As organisms become larger and multicellular, three things happen together:

- SA:V falls, so the outer body surface is too small relative to the volume of respiring tissue.
- Distances from the surface to the innermost cells become too great for diffusion to be fast enough.
- Many larger animals are more active and (in mammals and birds) maintain a high body temperature, so their metabolic rate and oxygen demand per unit volume are high.

The solution has two parts. **Specialised gas exchange surfaces** (gills, lungs, tracheal systems, leaves) provide a very large area with a thin barrier and a maintained gradient. A **mass transport system** (blood pumped by a heart, or xylem and phloem in plants) then moves substances in bulk flow over long distances between exchange surfaces and tissues, so that diffusion is only needed over the last short distance into each cell. Mass transport also helps to keep concentration gradients steep at the exchange surface by carrying absorbed substances away.

Exam Tip: Do not write that a large organism has a "smaller surface area". It has a larger surface area in absolute terms but a *smaller surface area to volume ratio*. Always use the word "ratio" and link it to the demand of the volume for oxygen, and to the diffusion distance.

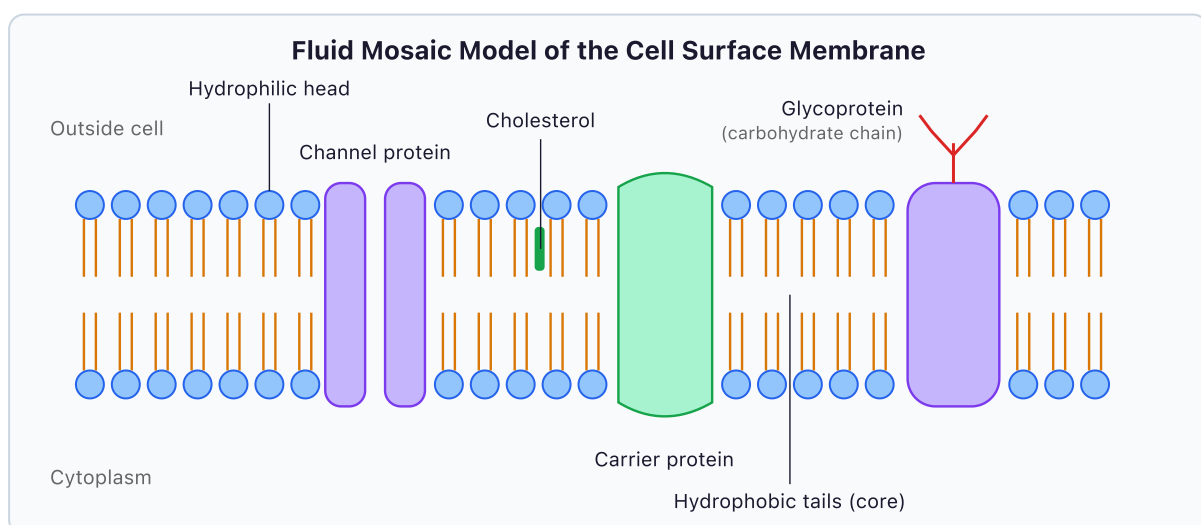
4.2 Cell Transport Mechanisms

4.2(i) The Fluid Mosaic Model

The cell surface membrane is a **phospholipid bilayer** about 7–10 nm thick. Each phospholipid has a hydrophilic phosphate head and two hydrophobic fatty acid tails. In water the molecules arrange themselves with heads facing the watery cytoplasm and tissue fluid, and tails pointing inwards, creating a hydrophobic core that most water-soluble substances cannot easily cross.

The model is called **fluid** because individual phospholipids and many proteins can move sideways within their layer, and **mosaic** because proteins of different shapes and sizes are scattered through the bilayer, like tiles in a mosaic.

- **Intrinsic (integral) proteins** span the bilayer. They include **channel proteins** (water-filled pores for particular ions or water) and **carrier proteins** (which bind a specific molecule and change shape to move it across).
- **Extrinsic (peripheral) proteins** sit on one surface; some act as enzymes or anchor the membrane to the cytoskeleton.
- **Glycoproteins** and **glycolipids** have carbohydrate chains on the outer surface. They act as receptors (for hormones, for example) and as recognition sites (antigens) in cell signalling.
- **Cholesterol** sits between the phospholipid tails in animal cells. It regulates fluidity, making the membrane more stable at higher temperatures and preventing it becoming too rigid at lower temperatures, and it reduces the leakage of water-soluble molecules.



4.2(ii) Passive Transport

Passive processes rely on the kinetic energy of the particles themselves. Net movement is always down a gradient, and the cell does not supply energy from ATP.

Diffusion: The net movement of molecules or ions from a region of higher concentration to a region of lower concentration, as a result of their random movement. It continues until the concentration is equal throughout (dynamic equilibrium).

Simple diffusion through the bilayer is possible for small, non-polar or lipid-soluble molecules such as O_2 , CO_2 and steroid hormones, which can dissolve in the hydrophobic core.

Facilitated diffusion is still passive and down a concentration gradient, but it requires membrane proteins:

- **Channel proteins** form hydrophilic pores. Each is specific to a particular ion (for example Na^+ , K^+ or Cl^-) or to water (aquaporins). Many are *gated*, opening only in response to a voltage change or the binding of a chemical.
- **Carrier proteins** bind a specific molecule, such as glucose, on one side of the membrane. Binding causes a change in the protein's shape that releases the molecule on the other side. Because there is a limited number of carriers, the rate levels off at high concentrations when all carriers are occupied.

Osmosis: The net movement of water molecules from a region of higher water potential to a region of lower water potential through a partially permeable membrane.

Water potential (ψ) is a measure of the tendency of water to move out of a solution. It is measured in kilopascals (kPa). Pure water at atmospheric pressure has the highest possible water potential, defined as 0 kPa. Dissolving solutes lowers it, so all solutions have negative water potentials; the more concentrated the solution, the more negative ψ becomes.

$$\text{Water potential} = \text{turgor pressure} + \text{osmotic potential} \quad \psi = P + \pi$$

- **Osmotic potential (π)** is the contribution of dissolved solutes. It is zero for pure water and negative for any solution.
- **Turgor pressure (P)** is the pressure of the cell contents pushing out against the cell wall. It is usually positive in a plant cell that has taken in water, and zero in a flaccid cell.

Worked example: A plant cell has an osmotic potential of -600 kPa and a turgor pressure of 250 kPa. Its water potential is $\psi = 250 + (-600) = -350$ kPa. If it is placed in a solution with $\psi = -700$ kPa, water leaves the cell by osmosis, because water moves from the higher (less negative) water potential, -350 kPa, to the lower one, -700 kPa. The cell loses turgor and, if enough water is lost, the cytoplasm pulls away from the wall (**plasmolysis**). In a solution with $\psi = 0$ kPa (pure water) the cell gains water until the rising wall pressure makes ψ inside equal 0 kPa; the cell is then fully turgid. An animal cell has no wall to resist this, so in pure water it swells and may burst (**lysis**); red blood cells in a strong solution shrink and become crenated.

Exam Tip: In osmosis answers use the phrase “water potential”, not “water concentration”, and always say which way the gradient runs (“from a higher water potential to a lower water potential”). Remember that -200 kPa is a *higher* water potential than -500 kPa. Osmosis is a special case of passive movement of *water*; do not describe solutes as moving by osmosis.

4.2(iii) How the Properties of Molecules Affect Their Transport

Property	Effect on crossing the membrane	Examples
Small and non-polar / lipid-soluble	Dissolve in the hydrophobic core and cross by simple diffusion	O_2 , CO_2 , steroid hormones, ethanol
Small and polar	Cross the bilayer only slowly; movement is much faster through specific channel proteins	Water (through aquaporins), urea
Charged (ions)	Repelled by the hydrophobic core and surrounded by a shell of water molecules; need channel or carrier proteins	Na^+ , K^+ , Cl^- , Ca^{2+}
Larger polar molecules	Too large and too polar to pass between phospholipids; need carrier proteins (facilitated diffusion or active transport)	Glucose, amino acids
Very large molecules and particles	Too large for any protein pathway; moved in vesicles	Proteins such as enzymes and hormones, bacteria taken in by phagocytes

4.2(iv) Endocytosis and Exocytosis

Large molecules and particles are moved in membrane-bound **vesicles**. Both processes need energy from ATP, because the cytoskeleton has to move the membrane and vesicles.

- **Endocytosis** (bulk transport into the cell): the cell surface membrane folds inwards around the material, and the edges fuse to pinch off a vesicle inside the cytoplasm. *Phagocytosis* is the uptake of solid particles, such as a bacterium taken in by a neutrophil; *pinocytosis* is the uptake of droplets of fluid.
- **Exocytosis** (bulk transport out of the cell): vesicles, often budded from the Golgi apparatus, move to the cell surface, their membrane fuses with the cell surface membrane, and the contents are released outside. Secretion of digestive enzymes, insulin and neurotransmitters all use exocytosis.

Core Practical 5 (CP5): The effect of temperature on beetroot membrane permeability.

Beetroot cells hold a red pigment (betalain) in their vacuoles. Discs heated at different temperatures release pigment into the surrounding water as the membranes are damaged, and the colour of the solution is measured with a colorimeter. For the method, safety, analysis and sources of error, see [Core Practicals: CP5](#).

Interpreting CP5 results: absorbance stays fairly low at moderate temperatures, then rises steeply once the temperature is high enough to *denature membrane proteins*, leaving gaps in the membrane. Higher temperatures also give the phospholipids more kinetic energy, so the bilayer becomes more fluid and leakier. Very low (freezing) temperatures can also increase leakage, because ice crystals can pierce membranes.

Core Practical 6 (CP6): Determine the water potential of plant cells. Pieces of plant tissue of equal size are placed in a range of solutions, and the percentage change in mass is plotted against concentration. The concentration giving no change in mass is the one whose water potential equals that of the tissue. See [Core Practicals: CP6](#).

Maths in CP6: percentage change in mass = $(\text{final mass} - \text{initial mass}) / \text{initial mass} \times 100$. Percentage change is used, rather than absolute change, so that samples with slightly different starting masses can be compared fairly. The line of best fit crosses the x-axis at the point of zero change (the intercept), which is read off and converted to a water potential.

4.2(v) Active Transport

Active transport: The movement of molecules or ions across a membrane against their concentration gradient (from lower to higher concentration), using carrier proteins and energy released by the hydrolysis of ATP.

1. The molecule or ion binds to a specific site on a carrier protein (often called a pump).
2. ATP binds to the carrier and is hydrolysed to ADP and inorganic phosphate (P_i). The phosphate attaches to the carrier protein.
3. This causes the carrier to change shape, moving the molecule to the other side of the membrane and releasing it.
4. The phosphate is released and the carrier returns to its original shape, ready to repeat the cycle.

A familiar example is the **sodium–potassium pump**, which moves 3 Na^+ out of the cell and 2 K^+ in for each ATP hydrolysed. Active transport is also used in root hair cells to take up mineral ions from the soil, and in the loading of sucrose into phloem (4.7(v)).

Because it depends on ATP from respiration, active transport is slowed or stopped by anything that reduces respiration: lack of oxygen, low temperature, or respiratory inhibitors such as cyanide. Cells that carry out a lot of active transport have **many mitochondria** and often a large membrane area (for example microvilli).

4.2(vi) ATP as the Energy Currency

ATP (adenosine triphosphate) is a nucleotide made of the base adenine, the pentose sugar ribose and three phosphate groups.

- **Phosphorylation of ADP** ($ADP + P_i \rightarrow ATP + \text{water}$) is a condensation reaction catalysed by ATP synthase. It *requires* energy, which comes from respiration (or from light in photosynthesis).
- **Hydrolysis of ATP** ($ATP + \text{water} \rightarrow ADP + P_i$) is catalysed by ATP hydrolase (ATPase) and *releases* energy, about 30.5 kJ per mole under standard conditions.



ATP is an **accessible** energy supply because it releases energy in a single-step reaction, in small, manageable amounts, exactly where it is needed. It is soluble and moves easily within the cell, and it is rapidly regenerated. It is used for active transport, muscle contraction, endocytosis and exocytosis, and the synthesis of large molecules. The inorganic phosphate released can also be attached to other molecules (phosphorylation), making them more reactive. ATP is not a long-term store: cells hold only a small amount and remake it continuously.

Exam Tip: Never write that ATP hydrolysis “produces energy” or that bonds “store energy”. Energy cannot be created. Say that hydrolysis of ATP *releases* energy, and that energy is *required* to phosphorylate ADP. Breaking any bond needs energy; the net release comes from the new bonds formed in the products.

4.3 Gas Exchange

4.3(i) Adaptations for Gas Exchange in Insects, Fish and Mammals

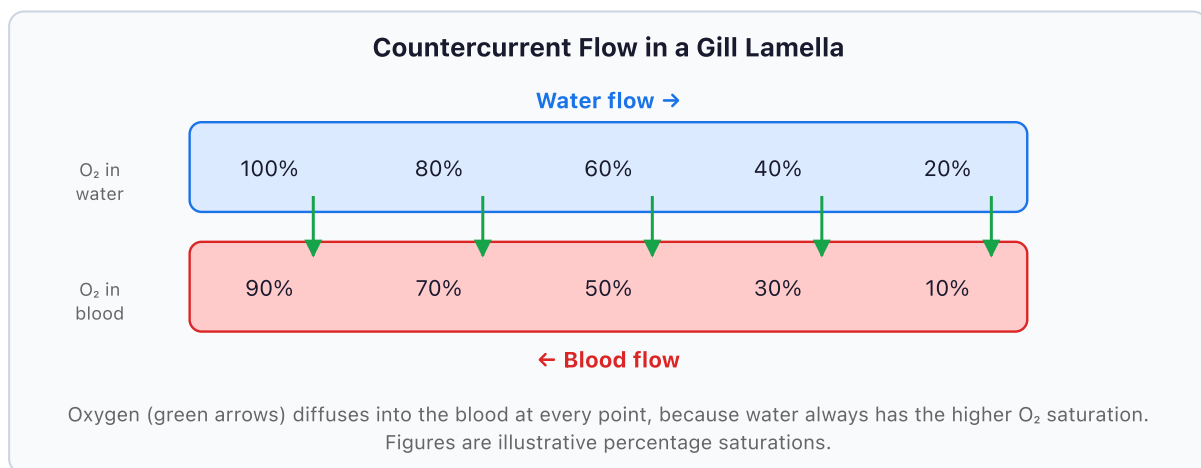
Insects have a hard exoskeleton covered with a waxy cuticle that reduces water loss but blocks gas exchange through the body surface. Instead, air is piped directly to the tissues through a **tracheal system**:

- **Spiracles** are pores along the sides of the thorax and abdomen. They can be closed by valves, reducing water loss when the insect is inactive.
- **Tracheae** are tubes running from the spiracles, held open by rings of chitin so they do not collapse.
- **Tracheoles** are very fine branches, each a single cell, that reach into the tissues and end right next to or within respiring cells, especially muscle. Oxygen diffuses along them and straight into the cells, so no blood transport of oxygen is needed.
- The ends of the tracheoles contain fluid. During intense activity, muscle cells respire anaerobically and produce lactate, which lowers the water potential of the cells. Water moves out of the tracheoles into the cells by osmosis, so air moves further along the tracheoles and gases diffuse faster through air than through water.
- Larger, active insects ventilate the system by **pumping movements of the abdomen**, which squeeze air in and out of the tracheae (and air sacs) to maintain steep gradients.

Core Practical 7 (CP7): Dissecting an insect to reveal its gas exchange system. Includes locating the spiracles, exposing the silvery air-filled tracheae and viewing chitin rings and tracheoles under the microscope, with attention to safe and ethical use of the specimen (humanely killed insects from a supplier, the minimum number, allergy precautions). See [Core Practicals: CP7](#).

Bony fish live in water, which holds far less dissolved oxygen than air and is denser and more viscous, so ventilation takes more energy. Their gills are highly adapted:

- Four pairs of gills, each with many thin **gill filaments** covered in **gill lamellae**, give a very large surface area. The water supports the filaments and keeps them apart.
- The lamellae have a thin epithelium and a dense capillary network, so the diffusion distance between water and blood is short.
- Water is pumped continuously over the gills in one direction: the mouth opens and the floor of the mouth (buccal cavity) lowers to draw water in, then the mouth closes, the floor rises, and water is forced over the gills and out under the **operculum** (gill cover).
- **Countercurrent flow:** blood in the lamellae flows in the opposite direction to the water flowing over them. Blood always meets water with a higher oxygen concentration than itself, so a diffusion gradient is maintained along the whole length of the lamella. Up to about 80% of the dissolved oxygen can be extracted. With parallel (concurrent) flow, the blood and water would reach equilibrium partway along, and at most about 50% could be taken up.



Mammals exchange gases in the lungs:

- Air passes down the trachea, bronchi and bronchioles to hundreds of millions of **alveoli**, which give a total surface area of the order of 70 m² in an adult human. The trachea and bronchi are held open by

rings of cartilage and lined with ciliated epithelium and mucus-secreting goblet cells.

- The alveolar wall and capillary wall are each a single layer of flattened (squamous) epithelial cells, so the diffusion distance between air and blood is less than 1 μm .
- Each alveolus is surrounded by a dense capillary network. Blood flow constantly removes oxygenated blood and brings deoxygenated blood, and **ventilation** constantly replaces alveolar air, so steep concentration gradients of O_2 and CO_2 are maintained.
- Capillaries are narrow, so red blood cells are squeezed against the wall, slowing them and shortening the diffusion distance.
- Elastic fibres let alveoli stretch on inhalation and recoil to help expel air; a film of **surfactant** lowers surface tension so alveoli do not collapse.

Ventilation: contraction of the external intercostal muscles and the diaphragm increases the volume of the thorax, lowering the pressure inside below atmospheric pressure so air flows in. When these muscles relax, the lungs recoil elastically, pressure rises and air flows out.

Feature	Insect	Bony fish	Mammal
Exchange surface	Tracheole walls	Gill lamellae	Alveolar walls
Large surface area from	Very many branching tracheoles	Many filaments and lamellae	Hundreds of millions of alveoli
Short diffusion distance	Tracheoles reach each respiring cell	Thin lamella epithelium	One-cell-thick alveolus and capillary walls
Gradient maintained by	Abdominal pumping; O_2 used by cells	One-way water flow; countercurrent flow	Ventilation; blood flow
Role of blood in O_2 transport	None (air reaches cells)	Yes	Yes

4.3(ii) Gas Exchange in Flowering Plants

Plants respire all the time and photosynthesise in the light. They have no ventilation mechanism and no blood; gases diffuse through air spaces. Leaves are thin and flat, so no cell is far from the air.

- **Stomata** are pores, mostly in the lower epidermis, each surrounded by a pair of **guard cells**. When guard cells take up K^+ ions (by active transport), their water potential falls, water enters by osmosis and they become turgid. The inner wall next to the pore is thicker and less flexible than the outer wall, so the cells bow apart and the pore opens. When K^+ leaves, water follows, the guard cells become flaccid and the pore closes. Stomata usually open in the light and close in the dark or when the plant is short of water.
- **Gas exchange surfaces in the leaf:** the **spongy mesophyll** has large **air spaces** between loosely packed cells, giving a very large internal surface of moist cell walls. Gases dissolve in the film of water on these walls and diffuse into or out of the cells. The palisade mesophyll cells are also in contact with air spaces.
- In the light, photosynthesis uses CO_2 faster than respiration produces it, so CO_2 diffuses in and O_2 diffuses out. In the dark, only respiration occurs, and the net direction reverses.
- **Lenticels** are areas of loosely packed cells in the bark of woody stems (and on some roots and fruits). The gaps between the cells let oxygen reach the living tissues beneath the impermeable cork, and CO_2 leave.

Exam Tip: Plants carry out *both* respiration and photosynthesis in the light. A common mistake is to say “plants photosynthesise in the day and respire at night”. Also note the trade-off at stomata: open stomata allow CO_2 in, but also allow water vapour to be lost by transpiration.

4.4 Circulation

4.4(i) Structure of the Heart and Blood Vessels

The mammalian heart is made of **cardiac muscle** and has four chambers. The right side pumps deoxygenated blood to the lungs; the left side pumps oxygenated blood to the rest of the body. A wall called the **septum** keeps the two sides separate.

- **Atria** (upper chambers) have thin walls; they receive blood from veins and pump it only a short distance into the ventricles.
- **Ventricles** (lower chambers) have thicker walls. The left ventricle wall is much thicker than the right, because it must generate a higher pressure to push blood all round the body, whereas the right ventricle only pumps to the nearby lungs.

- **Vessels:** the vena cava brings blood from the body to the right atrium; the pulmonary artery takes blood from the right ventricle to the lungs; the pulmonary vein returns blood from the lungs to the left atrium; the aorta carries blood from the left ventricle to the body. The **coronary arteries** branch from the aorta and supply the heart muscle itself.
- **Valves** keep blood flowing one way. The **atrioventricular (AV) valves** (tricuspid on the right, bicuspid or mitral on the left) lie between atria and ventricles, and are held by tendinous cords (chordae tendineae) that stop them turning inside out. **Semilunar valves** lie at the bases of the aorta and pulmonary artery.

Vessel	Structure	How structure suits function
Artery	Thick wall with much elastic tissue and smooth muscle; narrow lumen; folded endothelium; tough outer collagen layer	Withstands high pressure from the heart. Elastic tissue stretches during systole and recoils during diastole, smoothing flow and maintaining pressure.
Arteriole	More smooth muscle, less elastic tissue	Smooth muscle contracts or relaxes to control blood flow into particular capillary beds.
Capillary	Wall is a single layer of endothelium; lumen about the width of one red blood cell; gaps between endothelial cells	Very short diffusion distance; large total surface area; slow flow allows time for exchange; gaps allow tissue fluid to form.
Vein	Thin wall with little elastic tissue and muscle; wide lumen; valves	Blood is at low pressure, so a thick wall is not needed; the wide lumen reduces resistance; valves prevent backflow. Contraction of surrounding skeletal muscles squeezes the veins and helps return blood.

4.4(ii) Double and Single Circulation

In a **single circulation**, as in bony fish, blood passes through the heart once per circuit: heart → gills → body → heart. The narrow gill capillaries cause a large fall in pressure, so blood reaches the body tissues at low pressure and flows slowly.

Mammals have a **double circulation**: blood passes through the heart twice per circuit. The **pulmonary circulation** takes blood from the right ventricle to the lungs and back to the left atrium; the **systemic**

circulation takes blood from the left ventricle to the body and back to the right atrium. The advantages are:

- After losing pressure in the lung capillaries, blood returns to the heart and is **pumped again at high pressure** to the body. This gives rapid delivery of oxygen and glucose to tissues, which supports the high metabolic rate of an active endotherm.
- The pulmonary circulation can run at a **lower pressure**, which avoids damaging the delicate lung capillaries and reduces the amount of tissue fluid forming in the lungs.
- **Oxygenated and deoxygenated blood are kept apart**, so the blood sent to the body is as fully oxygenated as possible and the concentration gradients at the lungs and tissues are maximised.

4.4(iii) The Cardiac Cycle

One cardiac cycle is one complete heartbeat. At a heart rate of 75 beats per minute each cycle takes $60 / 75 = 0.8$ s. Blood always moves from higher to lower pressure, and valves open or close according to the pressures on either side of them.

1. **Atrial systole** (about 0.1 s): the atria contract, raising atrial pressure and pushing the remaining blood through the open AV valves into the ventricles. The ventricles are relaxed and already mostly full from passive filling.
2. **Ventricular systole** (about 0.3 s): the ventricles contract, starting from the apex (the bottom of the heart) and moving upwards. Ventricular pressure rises above atrial pressure and the AV valves close, preventing backflow into the atria. For a very short time all valves are shut and pressure rises steeply with no change in volume. When ventricular pressure exceeds the pressure in the aorta and pulmonary artery, the semilunar valves open and blood is ejected.
3. **Diastole** (about 0.4 s): the ventricles relax and their pressure falls. As it drops below the pressure in the arteries, the semilunar valves close, preventing backflow into the ventricles. When ventricular pressure falls below atrial pressure, the AV valves open. Blood returning in the veins flows into the atria and passively on into the ventricles.

4.4(iv) Myogenic Stimulation of the Heart

Cardiac muscle is **myogenic**: it contracts and relaxes rhythmically without needing impulses from nerves. Nerves and hormones only change the rate that the heart sets for itself.

1. The **sinoatrial node (SAN)**, a patch of specialised tissue in the wall of the right atrium, acts as the pacemaker. It spontaneously sends out waves of electrical excitation (depolarisation) at regular

intervals.

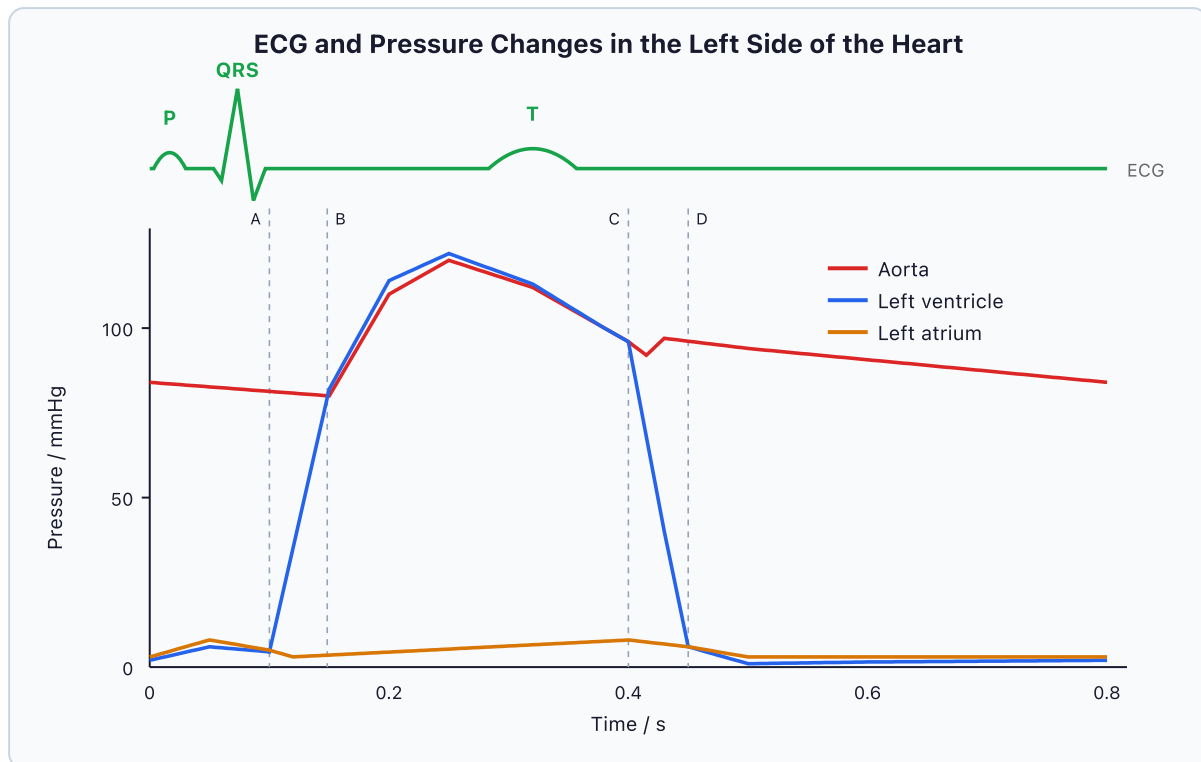
2. The wave spreads across both atria, causing them to contract (atrial systole).
3. A layer of non-conducting fibrous tissue between the atria and ventricles stops the wave passing straight into the ventricles. The only route is through the **atrioventricular node (AVN)**, which delays the impulse by about 0.1 s. The delay lets the atria finish emptying before the ventricles contract.
4. The AVN passes the excitation down the **bundle of His**, conducting fibres in the septum, to the apex of the heart.
5. The bundle branches into **Purkinje fibres** in the ventricle walls, which carry the excitation upwards through the ventricle muscle. The ventricles contract from the apex upwards, squeezing blood up and out into the arteries.

Exam Tip: Describe the SAN as sending out a “wave of electrical excitation” or “wave of depolarisation”, not a nerve impulse, and say it is in the *right atrium*. Explain *why* there is a delay at the AVN (so the atria empty before the ventricles contract) and *why* the ventricles contract from the apex (to push blood upwards towards the arteries).

4.4(v) Interpreting ECG Traces and Pressure Changes

An **electrocardiogram (ECG)** records the electrical activity of the heart using electrodes on the skin. A normal trace has three features:

- **P wave:** depolarisation of the atria, leading to atrial systole.
- **QRS complex:** depolarisation of the ventricles, leading to ventricular systole. It is large because the ventricles have much more muscle than the atria. (Atrial repolarisation happens at the same time and is hidden.)
- **T wave:** repolarisation of the ventricles, as they return to rest before diastole.



Reading the pressure graph (points A–D on the diagram):

- **A – AV valve closes:** ventricular pressure rises above atrial pressure at the start of ventricular systole, just after the QRS complex.
- **B – semilunar valve opens:** ventricular pressure rises above aortic pressure and blood is ejected. Between A and B all valves are closed.
- **C – semilunar valve closes:** ventricular pressure falls below aortic pressure after the T wave. The small dip and rebound in the aortic trace here (the *dicrotic notch*) is caused by the valve closing and the elastic recoil of the aorta.
- **D – AV valve opens:** ventricular pressure falls below atrial pressure, and the ventricle starts to fill again. Between C and D all valves are closed.

Aortic pressure never falls close to zero, because the elastic aortic wall recoils during diastole. The right ventricle follows the same pattern but reaches a much lower peak pressure, since it pumps only to the lungs.

Heart rate from an ECG: measure the time between two consecutive R peaks (the R–R interval) and use $\text{heart rate} = 60 / \text{R–R interval in seconds}$. For an interval of 0.8 s, $\text{heart rate} = 60 / 0.8 = 75$ beats per minute. On ECG paper, first work out the time represented by each square from the paper speed. You may also be shown abnormal traces: a resting rate above 100 beats per minute (tachycardia) or below 60 (bradycardia), or an irregular trace with no clear P waves (atrial fibrillation).

Exam Tip: When asked to “explain” a valve event from a graph, name the two pressures being compared and which is higher, for example “the semilunar valve closes because the pressure in the ventricle falls below the pressure in the aorta”. Do not say the valves open because the chamber contracts; it is always the pressure difference.

4.4(vi) Structure of Blood

Blood is made of a liquid, **plasma**, carrying **cells** and cell fragments. Plasma is mostly water with dissolved glucose, amino acids, mineral ions, hormones, urea, CO₂ (mainly as hydrogencarbonate ions) and **plasma proteins** such as albumin, fibrinogen and antibodies.

Cell type	Appearance	Main role
Erythrocyte (red blood cell)	Biconcave disc about 7 µm across; no nucleus, mitochondria or ER; packed with haemoglobin	Carries O ₂ (and some CO ₂). The biconcave shape gives a large SA:V and short diffusion distance; lacking a nucleus leaves more room for haemoglobin; being flexible lets it squeeze through capillaries.
Neutrophil	Multi-lobed nucleus; granular cytoplasm; the most numerous white cell	Phagocytosis of bacteria and other pathogens; first to arrive at infections
Eosinophil	Usually a two-lobed nucleus; cytoplasm with large granules that stain red with eosin	Defence against parasites such as worms; involved in allergic and inflammatory reactions
Monocyte	The largest white cell; large kidney-shaped or bean-shaped nucleus	Leaves the blood and becomes a macrophage, which carries out phagocytosis and presents antigens
Lymphocyte	Small cell with a large round nucleus that almost fills it; little cytoplasm	Specific immune response: B cells make antibodies; T cells have several roles, including killing infected cells

Neutrophils, eosinophils and monocytes are all **leucocytes** (white blood cells), as are lymphocytes. **Platelets** are small fragments of larger cells, with no nucleus, and are involved in clotting (4.4(viii)).

4.4(vii) Functions of Blood

- **Transport:** oxygen from the lungs to respiring tissues and CO₂ back to the lungs; glucose, amino acids and other products of digestion from the gut; urea from the liver to the kidneys; hormones from endocrine glands to target organs; heat from active organs around the body.
- **Defence:** white blood cells destroy pathogens by phagocytosis and by the specific immune response (antibodies); platelets and plasma proteins form clots that seal wounds, stopping blood loss and the entry of pathogens.
- **Formation of tissue fluid and lymph:** plasma leaks out of capillaries to form tissue fluid, which bathes the cells, and excess tissue fluid drains away as lymph (4.6).

4.4(viii) Blood Clotting

When a blood vessel is damaged, a series of events rapidly seals the wound:

1. Damage exposes **collagen** fibres in the vessel wall. **Platelets** stick to the collagen and to each other, forming a **platelet plug**.
2. The platelets and damaged tissue release **clotting factors**, including **thromboplastin** (an enzyme).
3. Thromboplastin, together with calcium ions (Ca²⁺) as an essential cofactor, catalyses the conversion of the inactive plasma protein **prothrombin** into the active enzyme **thrombin**.
4. Thrombin catalyses the conversion of the soluble plasma protein **fibrinogen** into insoluble **fibrin**.
5. Fibrin forms a mesh of long fibres over the wound. It traps more platelets and red blood cells to form a **clot**, which dries into a scab.

prothrombin → thrombin (catalysed by thromboplastin, needs Ca²⁺) | fibrinogen (soluble) →
fibrin (insoluble) (catalysed by thrombin)

Because each step is catalysed by an enzyme, a small trigger is greatly amplified into a rapid response. Clotting factors circulate in inactive forms, so clots only form where they are needed.

Exam Tip: Keep the names straight: *thromboplastin* (released by platelets) acts on *prothrombin* to form *thrombin*; thrombin then acts on *fibrinogen* to form *fibrin*. Always state that fibrinogen is

soluble and fibrin is insoluble, as marks are often given for this.

4.4(ix) Atherosclerosis

Atherosclerosis: A disease of arteries in which fatty deposits and fibrous tissue (plaques, or atheromas) build up in the artery wall beneath the endothelium, narrowing the lumen and making the wall less elastic.

Stages:

1. The **endothelium** lining an artery is damaged, for example by high blood pressure or by chemicals in tobacco smoke.
2. An **inflammatory response** follows. White blood cells (monocytes, which become macrophages) move into the artery wall.
3. **Cholesterol** (carried in LDLs), along with the macrophages, accumulates under the endothelium, forming a fatty streak. Macrophages that take up lipid become foam cells.
4. Calcium salts, fibrous tissue and smooth muscle cells build up at the site, forming a hard **plaque (atheroma)**. The artery wall loses elasticity and the lumen narrows.
5. The narrower lumen raises blood pressure, which can damage the endothelium further, so the process is self-reinforcing (positive feedback).
6. The rough surface of the plaque, or a plaque that ruptures, can trigger the clotting sequence (4.4(viii)), forming a **thrombus**. A thrombus may block the artery where it forms, or break off (an **embolus**) and block a smaller artery elsewhere.

Effects on health:

- Narrowed coronary arteries reduce oxygen supply to heart muscle, causing chest pain on exertion (**angina**). A complete blockage causes a **myocardial infarction** (heart attack): the heart muscle beyond it cannot respire aerobically and begins to die.
- A blockage of an artery supplying the brain causes a **stroke**, in which brain tissue is starved of oxygen.
- Weakened artery walls can bulge (an **aneurysm**) and may burst, causing a haemorrhage.
- Reduced blood supply to the legs can cause pain and, in severe cases, tissue death.

Factors that increase the risk:

Risk factor	How it increases the risk
Genetics (family history)	Some inherited alleles lead to high blood cholesterol or high blood pressure
Age	Arteries become less elastic and plaques accumulate over many years
Sex	Men have a higher risk than women before the menopause
High blood pressure	Damages the endothelium, starting the process
Smoking	Chemicals damage the endothelium; nicotine raises heart rate and blood pressure; carbon monoxide reduces O ₂ carriage, so the heart works harder
Diet high in saturated fat and salt	Saturated fat tends to raise blood LDL cholesterol; salt tends to raise blood pressure
Physical inactivity and obesity	Associated with high blood pressure, high blood lipids and type 2 diabetes
Diabetes	High blood glucose damages the endothelium

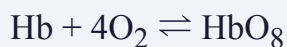
Exam Tip: Atherosclerosis happens in the *wall* of an artery, under the endothelium, not as fat sticking to the inside of the lumen. Start the sequence with endothelial damage. Keep atherosclerosis (the plaque) separate from thrombosis (the clot) and from the outcome (angina, heart attack or stroke).

4.5 Transport of Gases in the Blood

4.5(i) Haemoglobin and the Bohr Effect

Haemoglobin (Hb) is a globular protein with **quaternary structure**: four polypeptide chains (two α and two β in adults), each folded around a **haem group** containing an iron ion (Fe^{2+}). Each haem can bind

one O₂ molecule, so one haemoglobin molecule carries up to four O₂ molecules, forming oxyhaemoglobin:



The binding of oxygen is **cooperative**. When the first O₂ binds, the shape of the molecule changes slightly, making it easier for the second and third to bind. The last one binds less readily because few sites are left. This explains the S-shaped curve in 4.5(ii). Haemoglobin is soluble, because its hydrophilic R-groups face outwards, so red blood cells can hold a lot of it.

Carbon dioxide transport: CO₂ is carried in three ways: a small proportion dissolved in plasma, some bound to haemoglobin as carbaminohaemoglobin, and most as hydrogencarbonate ions. In red blood cells, the enzyme **carbonic anhydrase** catalyses the reaction of CO₂ with water to form carbonic acid, which dissociates into H⁺ and HCO₃⁻. The HCO₃⁻ ions diffuse out into the plasma, and chloride ions move in to balance the charge (the **chloride shift**). The H⁺ ions bind to haemoglobin, forming haemoglobinic acid, which buffers the blood pH and causes oxyhaemoglobin to release its oxygen.

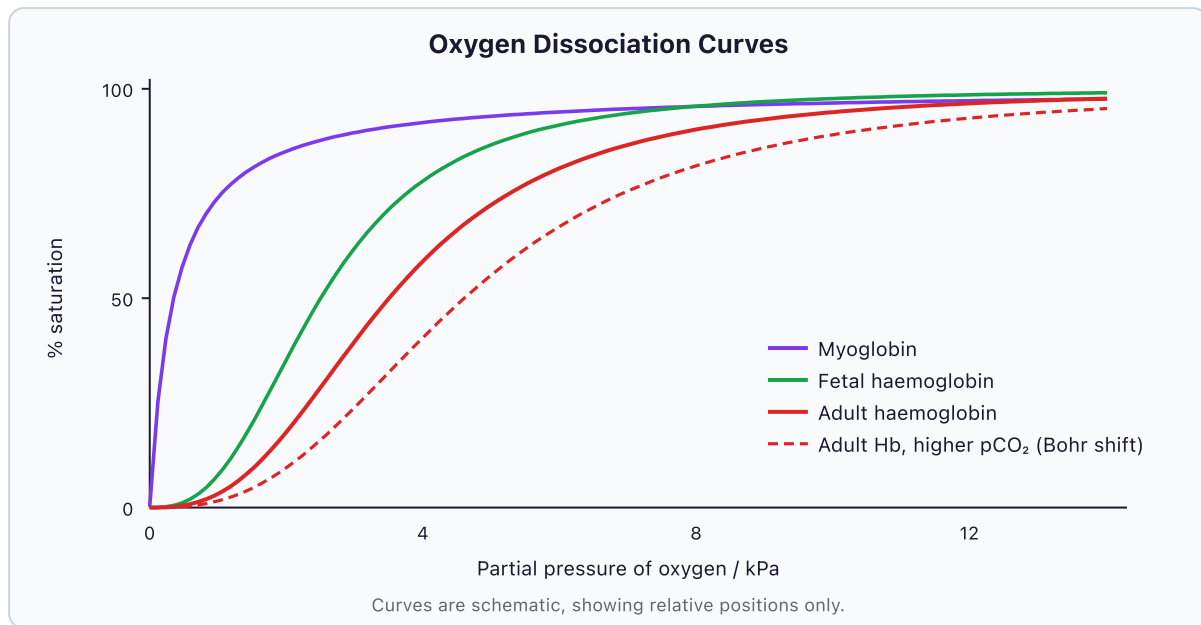


The Bohr effect: A higher partial pressure of carbon dioxide lowers the affinity of haemoglobin for oxygen, shifting the oxygen dissociation curve to the right. Respiring tissues produce CO₂, so haemoglobin unloads more oxygen exactly where it is most needed.

The effect works because CO₂ forms H⁺ ions (lowering pH), and H⁺ binding changes the shape of haemoglobin so that it holds oxygen less tightly. In the lungs, where CO₂ is low, the curve is effectively further left and haemoglobin loads oxygen more readily.

4.5(ii) The Oxygen Dissociation Curve

An oxygen dissociation curve plots the percentage saturation of haemoglobin with oxygen against the partial pressure of oxygen (pO₂, in kPa). Partial pressure measures how much of a gas is present in a mixture or dissolved in a liquid.



- **At high pO₂** (in the lungs, around 12–13 kPa) haemoglobin is almost fully saturated, so it *loads* oxygen. The curve is nearly flat here, so a modest fall in alveolar pO₂ hardly affects loading.
- **On the steep middle part** (roughly 2–6 kPa, typical of respiring tissues) a small fall in pO₂ causes a large fall in saturation, so a lot of oxygen is *unloaded* for a small change in conditions.
- **At low pO₂** the curve is shallow because it is hard for the first O₂ molecule to bind; once it has, cooperative binding produces the steep rise.
- **Shift to the right** (Bohr effect, also higher temperature): lower affinity, less saturation at any given pO₂, more oxygen released to tissues. This happens in actively respiring muscle.
- **Shift to the left**: higher affinity, loads more readily but unloads less readily.

Using the curve: the amount of oxygen released is the difference between the saturation in the lungs and the saturation in the tissue. If Hb is 97% saturated in the lungs and 25% saturated in a muscle, then $97 - 25 = 72$ percentage points of the oxygen carried have been released, or $72 / 97 \times 100 = 74\%$ of the oxygen that was loaded.

Exam Tip: Link the *shape* of the curve to cooperative binding, and the *position* of the curve to affinity. "Shifted to the right" means lower affinity and easier unloading. When comparing two curves, quote values at the same pO₂ read from the graph, and state the units (kPa and %).

4.5(iii) Haemoglobin and Myoglobin

Myoglobin is an oxygen-binding protein found in muscle cells, especially in the muscles of diving mammals and in slow (red) muscle fibres. It acts as an oxygen store.

Feature	Haemoglobin	Myoglobin
Structure	Four polypeptide chains; quaternary structure	One polypeptide chain; tertiary structure only
Haem groups	Four (each with Fe^{2+})	One (with Fe^{2+})
O_2 molecules bound	Up to four	One
Cooperative binding	Yes, giving an S-shaped (sigmoid) curve	No, giving a hyperbolic curve
Oxygen affinity	Lower; curve to the right	Much higher; curve far to the left
Bohr effect	Yes	No significant effect
Location	Red blood cells	Muscle cells
Function	Transports O_2 from lungs to tissues	Stores O_2 in muscle; releases it only at very low pO_2 , such as during intense exercise

Similarities: both are globular, water-soluble proteins; both contain haem with Fe^{2+} ; and both bind oxygen reversibly. Because myoglobin has the higher affinity, it takes oxygen from oxyhaemoglobin in muscle capillaries and keeps hold of it until the muscle's own pO_2 drops very low.

4.5(iv) Fetal Haemoglobin

A fetus gets its oxygen from its mother's blood in the **placenta**; the two bloodstreams do not mix, and oxygen diffuses across. By the time maternal blood reaches the placenta it has already given up some oxygen, so the pO_2 there is low. **Fetal haemoglobin** has a slightly different polypeptide composition (two γ chains in place of the two β chains) and a **higher affinity for oxygen** than adult haemoglobin, so its curve lies to the *left*.

- At the low pO_2 in the placenta, adult haemoglobin unloads oxygen while fetal haemoglobin loads it. Oxygen therefore moves from mother to fetus.
- The Bohr effect helps: the fetus gives CO_2 to the maternal blood, which shifts the mother's curve further to the right and promotes unloading.
- After birth the baby breathes air, and fetal haemoglobin is gradually replaced by adult haemoglobin over the first months, since a lower affinity is better for unloading in the tissues.

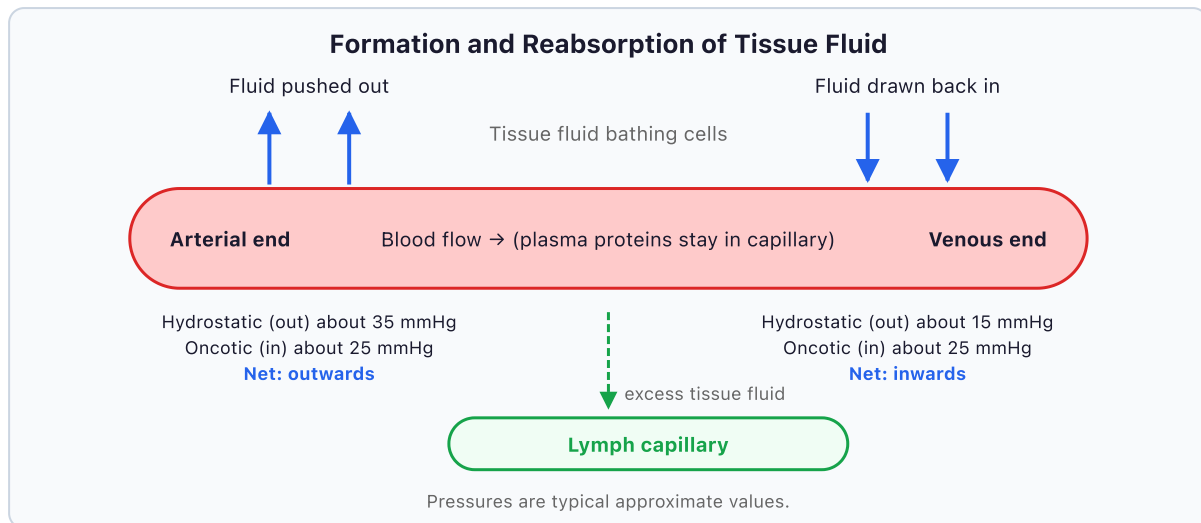
Exam Tip: Say that fetal haemoglobin has a *higher affinity* for oxygen "at the same partial pressure". Do not say fetal haemoglobin "takes" or "pulls" oxygen off the mother's haemoglobin. Explain it in terms of the maternal Hb dissociating and the fetal Hb associating at the low pO_2 of the placenta.

4.6 Transfer of Materials Between the Circulatory System and Cells

4.6(i) Formation and Reabsorption of Tissue Fluid

Cells are not in direct contact with blood. They are bathed in **tissue fluid**, which is formed from plasma and has a similar composition, except that it has almost no plasma proteins or red blood cells, as these are too large to pass through the capillary wall. Exchange between blood and cells happens through this fluid. Two opposing pressures decide which way fluid moves across the capillary wall:

- **Hydrostatic pressure** is the pressure of the blood pushing outwards on the capillary wall. It comes from the pumping of the heart and is highest at the arterial end of the capillary.
- **Oncotic pressure** is the osmotic effect of the plasma proteins (mainly albumin) that stay in the blood. They lower the water potential of the plasma, tending to draw water back into the capillary. It stays roughly constant along the capillary.



1. **At the arterial end**, hydrostatic pressure is greater than oncotic pressure, so there is a net outward force. Water and small solutes (glucose, amino acids, fatty acids, ions, O_2) are forced out through gaps between the endothelial cells, forming tissue fluid. This process is sometimes called pressure filtration.
2. Cells take what they need from the tissue fluid and release waste such as CO_2 and urea into it.
3. Hydrostatic pressure falls along the capillary because fluid has been lost and because of resistance to flow.
4. **At the venous end**, hydrostatic pressure is now lower than oncotic pressure, so there is a net inward force. Water moves back into the capillary by osmosis, carrying dissolved wastes with it.

If blood pressure is raised, more fluid is forced out and less returns, so tissue fluid builds up and the tissues swell (**oedema**). If plasma protein concentration falls, for example in severe protein deficiency, oncotic pressure is lower and less fluid is reabsorbed, again causing oedema.

4.6(ii) The Lymphatic System

Not all of the tissue fluid returns to the capillaries. The excess drains into blind-ended **lymph capillaries** and is then called **lymph**. Lymph capillaries join into larger lymph vessels that carry lymph slowly through lymph nodes (where lymphocytes can detect and respond to pathogens). The flow is driven by the squeezing of the vessels by surrounding skeletal muscles, with **valves** preventing backflow. Lymph is returned to the blood where the main lymph vessels empty into veins near the heart (the subclavian veins). The lymphatic system also transports lipids absorbed from the small intestine.

Exam Tip: When explaining tissue fluid, compare the two pressures at each end explicitly, e.g. “at the arterial end hydrostatic pressure is greater than oncotic pressure, so fluid is forced out”. Reabsorption at the venous end is by *osmosis*, because the plasma proteins have lowered the water potential of the blood.

4.7 Transport in Plants

4.7(i) Structure of Xylem and Phloem

In a flowering plant, xylem and phloem are grouped in **vascular bundles**. Xylem carries water and mineral ions upwards from the roots; phloem carries dissolved organic substances, mainly sucrose and amino acids, from where they are made or released (**sources**) to where they are used or stored (**sinks**), in either direction.

Feature	Xylem vessels	Phloem sieve tubes
Living or dead?	Dead when mature; no cytoplasm	Living sieve tube elements with a thin layer of cytoplasm but no nucleus, and few organelles
End walls	Lost completely, forming a continuous hollow tube	Sieve plates: end walls perforated with pores
Cell walls	Thickened with lignin, often in rings or spirals; pits allow sideways movement	Cellulose walls, not lignified
Associated cells	Supported by fibres and xylem parenchyma	Each has a companion cell linked by many plasmodesmata
Flow	Upwards only; passive (driven by transpiration)	Up or down; needs energy for loading

- **Xylem:** the lack of end walls and cytoplasm means water can flow as an unbroken column with little resistance. **Lignin** makes the walls waterproof and strong, preventing the vessels collapsing under the tension produced by transpiration, and helps support the plant. Lignin also helps water adhere to

the walls. **Pits** (unlignified areas) let water move sideways between vessels and into surrounding cells.

- **Phloem:** sieve tube elements have lost most of their organelles to leave more room for flow. **Companion cells** have a nucleus, dense cytoplasm and many mitochondria, providing ATP for the active loading of sucrose, and they support the metabolism of the sieve tube element through the plasmodesmata.

4.7(ii) Apoplastic and Symplastic Pathways

Water enters root hair cells by osmosis, because the cell sap has a lower water potential than the soil water (root hair cells actively take up mineral ions). It then crosses the root cortex to the xylem, moving down a water potential gradient, by two main routes:

- **Apoplastic pathway:** water moves through the cellulose **cell walls** and the spaces between cells, without crossing any membrane. It is drawn along by cohesion between water molecules as water moves into the xylem. This route offers little resistance and carries most of the water in the cortex.
- **Symplastic pathway:** water moves through the **cytoplasm** of cells, passing from cell to cell through **plasmodesmata** (strands of cytoplasm through the cell walls). It enters the first cell by osmosis across its membrane.

Some descriptions add a vacuolar pathway, in which water also passes through the vacuoles of cells.

At the **endodermis**, the layer of cells around the vascular tissue, the apoplastic pathway is blocked by the **Casparian strip**, a band of waterproof suberin in the cell walls. Water must now cross the cell surface membrane into the cytoplasm and continue by the symplastic route. This lets the plant control which mineral ions enter the xylem, since they must pass through membrane proteins. Endodermal cells actively transport mineral ions into the xylem, lowering its water potential so that water follows by osmosis. This creates **root pressure**, which makes a small contribution to moving water upwards.

4.7(iii) The Cohesion-Tension Model

1. Water evaporates from the moist walls of mesophyll cells into the air spaces of the leaf, and water vapour diffuses out through open stomata. This loss of water vapour is **transpiration**.
2. Mesophyll cells replace the lost water from neighbouring cells and from the xylem in the leaf veins, by osmosis and through the apoplast.
3. Water molecules are held together by hydrogen bonds (**cohesion**) and are attracted to the lignified xylem walls (**adhesion**). So as water leaves the top of the xylem, the whole continuous column is

pulled upwards, a pulling force called the **transpiration pull**.

4. The column is under **tension** (negative pressure). Water is drawn into the xylem in the roots to replace water moving up, and so the transpiration stream flows from roots to leaves.

Evidence for the model: tree trunks measurably shrink in diameter during the day when transpiration is fastest, because tension pulls the xylem walls inwards; if a xylem vessel is cut, air is drawn in rather than water leaking out; and once air enters and breaks the column (an air lock), water can no longer be pulled up that vessel.

Exam Tip: Transpiration is the loss of water *vapour* from the leaves, mainly through stomata; it is the driving force for the movement of water up the stem, not the movement itself. Use the words *cohesion* (water to water), *adhesion* (water to xylem wall), *tension* and *continuous column*.

4.7(iv) Factors Affecting the Rate of Transpiration

Factor	Effect on rate	Explanation
Temperature	Higher temperature increases rate	Water molecules have more kinetic energy, so evaporation from cell walls and diffusion of vapour are faster; warmer air can also hold more water vapour, so the gradient between leaf and air is steeper
Light	Brighter light increases rate (up to a point)	Stomata open in the light to allow CO ₂ in for photosynthesis, so more water vapour can diffuse out
Humidity	Higher humidity decreases rate	More water vapour in the air reduces the water vapour potential gradient between the air spaces in the leaf and the outside air
Air movement	More air movement (wind) increases rate	Moving air removes water vapour from just outside the stomata, keeping the diffusion gradient steep; still air lets a layer of humid air build up

Core Practical 8 (CP8): Investigate factors affecting water uptake by plant shoots using a potometer. A leafy shoot is cut and fitted into a potometer under water with no air leaks, and the

rate at which an air bubble moves along a capillary tube is measured under different conditions (for example with a fan, a lamp, or a plastic bag over the shoot). See [Core Practicals: CP8](#).

Potometer maths: the volume of water taken up equals the volume of the cylinder of capillary tube through which the bubble moved: $V = \pi r^2 \times \text{distance}$. For example, if the capillary has an internal diameter of 0.5 mm (so $r = 0.25$ mm) and the bubble moves 36 mm in 4 minutes, $V = \pi \times 0.25^2 \times 36 = 7.07 \text{ mm}^3$, and the rate of water uptake = $7.07 / 4 = 1.77 \text{ mm}^3 \text{ min}^{-1}$. A potometer measures water *uptake*, which is only an estimate of transpiration: a small amount of water is used by the plant for photosynthesis and to keep cells turgid.

4.7(v) The Mass-Flow Hypothesis for Phloem Transport

Movement of organic solutes in the phloem is called **translocation**. The mass-flow (pressure flow) hypothesis, proposed by Münch, explains it as follows:

1. **At the source** (for example a photosynthesising leaf), sucrose is actively loaded into the sieve tube elements. Companion cells use ATP to pump H^+ ions out; the H^+ ions then diffuse back in through cotransporter proteins, carrying sucrose with them.
2. This lowers the water potential in the sieve tube, so water enters by osmosis from the nearby xylem. This raises the **hydrostatic pressure** at the source end.
3. **At the sink** (for example a root, growing tip or developing fruit), sucrose is removed from the phloem and used in respiration or converted to starch for storage. This raises the water potential, so water leaves by osmosis and hydrostatic pressure is lower.
4. The pressure difference between source and sink pushes the sap along the sieve tubes by mass flow, from high to low pressure.

Strengths (evidence consistent with mass flow)	Weaknesses (observations it does not fully explain)
Sap flows out under pressure when phloem is cut, or from the mouthparts of an aphid feeding on a sieve tube, showing the contents are under positive pressure.	Sieve plates would be expected to slow mass flow a great deal, so it is unclear why they are there.
The sucrose concentration in phloem sap is higher near sources than near sinks.	Different substances in the same phloem have been found moving at different speeds, and in

	some cases in opposite directions, which a single bulk flow cannot explain.
Removing a ring of bark (and so the phloem) from a stem causes sugars to accumulate above the ring; radioactive tracers such as $^{14}\text{CO}_2$ supplied to a leaf show labelled sugars travelling from source to sinks.	Sucrose is delivered to sinks according to their demand, not simply to the region of lowest pressure.
Metabolic inhibitors or lack of oxygen stop translocation, consistent with the need for ATP in loading.	Companion cells have many mitochondria and the sieve tube itself is living, suggesting energy may be involved beyond loading, which the passive flow part of the model does not account for.
Rates of translocation are far faster than diffusion alone could achieve.	The model does not explain a role for the phloem proteins found in sieve tubes.

Exam Tip: An “evaluate” question on mass flow needs both sides and a conclusion, for example that the hypothesis explains most observations but not all, so it is still the best available model. Remember that the *loading* of sucrose is active, but the *flow* itself is passive, driven by the hydrostatic pressure gradient.

Command Words and Common Misconceptions

Command word	What Edexcel wants	Example from this topic
Describe	Give an account of what happens or what is shown, including data, without reasons	Describe the shape of the oxygen dissociation curve
Explain	Give reasons, using biological knowledge, for why something happens	Explain why the left ventricle wall is thicker than the right
Calculate	Work out a numerical answer, showing working and units	Calculate the rate of water uptake from potometer readings

Compare	Similarities and differences, both sides stated in each point	Compare the structure of xylem vessels and sieve tubes
Evaluate	Weigh up evidence for and against, then reach a judgement	Evaluate the mass-flow hypothesis
Deduce / Suggest	Use the information given to reach a conclusion, or apply knowledge to an unfamiliar situation	Suggest why a diving mammal has a high concentration of myoglobin

- **"Veins carry deoxygenated blood."** Veins carry blood towards the heart; the pulmonary vein carries oxygenated blood. Likewise, the pulmonary artery carries deoxygenated blood.
- **"Haemoglobin with a lower affinity is worse."** A lower affinity (curve to the right) is an advantage in active tissues, where more unloading is needed.
- **"The potometer measures transpiration."** It measures water uptake, which only approximates transpiration.

Key Terms Checklist

Check that you can define or explain each of these terms and link it to the spec statement it belongs to.

- **4.1** surface area to volume ratio; Fick's law; diffusion distance; mass transport system; specialised exchange surface
- **4.2** fluid mosaic model; phospholipid bilayer; intrinsic and extrinsic proteins; glycoprotein; cholesterol; diffusion; facilitated diffusion; channel protein; carrier protein; aquaporin; osmosis; water potential (ψ); turgor pressure (P); osmotic potential (π); plasmolysis; endocytosis; exocytosis; phagocytosis; vesicle; active transport; sodium–potassium pump; ATP; phosphorylation; hydrolysis; ATP hydrolase; ATP synthase
- **4.3** spiracle; trachea; tracheole; gill filament; lamella; operculum; countercurrent flow; alveolus; surfactant; ventilation; stoma; guard cell; spongy mesophyll; air space; lenticel
- **4.4** atrium; ventricle; septum; atrioventricular valve; semilunar valve; tendinous cords; coronary artery; artery; arteriole; capillary; vein; single circulation; double circulation; pulmonary and systemic circulation; cardiac cycle; systole; diastole; myogenic; sinoatrial node; atrioventricular node; bundle of His; Purkinje fibres; ECG; P wave; QRS complex; T wave; plasma; erythrocyte; leucocyte;

neutrophil; eosinophil; monocyte; lymphocyte; platelet; thromboplastin; prothrombin; thrombin; fibrinogen; fibrin; atherosclerosis; atheroma; endothelium; thrombus; embolus

- **4.5** haemoglobin; haem group; cooperative binding; partial pressure; oxygen dissociation curve; affinity; Bohr effect; carbonic anhydrase; chloride shift; myoglobin; fetal haemoglobin; placenta
- **4.6** tissue fluid; hydrostatic pressure; oncotic pressure; oedema; lymph; lymph capillary; lymph node
- **4.7** xylem vessel; lignin; pit; sieve tube element; sieve plate; companion cell; plasmodesmata; apoplastic pathway; symplastic pathway; endodermis; Casparian strip; root pressure; transpiration; cohesion; adhesion; tension; transpiration pull; potometer; translocation; source; sink; mass-flow hypothesis